

SOL.52 — Solar Commerce + Proposal Network

Architecture Blueprint · May 2026

Status: DEFERRED INFRASTRUCTURE — Reference Only

SOL.52 — Solar Commerce + Proposal Network Architecture

Companion to the prior **Instant Proposal + Quotation Architecture** plan. That plan defines *how a single org sells*. This plan defines *how many orgs sell together* on one platform — sellers, listings, inventory, commissions, fulfillment, and the Commerce Workspace.

STATUS: DEFERRED INFRASTRUCTURE — DO NOT IMPLEMENT YET

Decision (2026-05-19): This entire blueprint is **frozen as deferred infrastructure**. No phase below (E1–E7) is to be implemented in the current cycle.

Unlock condition — implementation may begin only when ALL of the following are true:

1. The Proposal OS has reached **production-quality maturity** — Phase D (Instant Proposal + Quotation) is fully shipped, the unified builder is stable on mobile + desktop, the canonical engine has zero open correctness bugs, and `proposal_pricing_snapshots` + `proposal_approval_events` audit flows are battle-tested with real customer data.
2. The current Proposal OS roadmap (E0–E13 from the prior Experience Blueprint, plus Phase D from the Instant Proposal + Quotation Architecture plan) is signed off as "done" by the product owner.
3. A go/no-go review explicitly references this document and re-confirms the architecture is still correct for the market reality at that future date.

Current priority — Proposal OS perfection. All design, engineering, and QA effort flows to the proposal engine, builder UX, residential/commercial parity, story modes, share flows, and intelligence (org-type defaults, AI quick suggestions). Marketplace, vendor onboarding, listings, orders, commissions, and Commerce Workspace are **out of scope** until the unlock condition is met.

What this document is for, in the meantime:

- A **preservation contract** — every choice below (single-tenant root, additive pricing-line fields, snapshot-on-send, append-only commission ledger, organization-mode flags, layered architecture) is the agreed long-term shape. Day-to-day Proposal OS work must remain *compatible* with it (no destructive moves that would block these phases later), but must not *anticipate* it with premature scaffolding.

- A **reference for safe seams** — when Proposal OS work touches organizations, `proposal_pricing.line_items`, `proposal_pricing_snapshots`, or `proposal_approval_events`, contributors should keep this document open to ensure additive-only changes (per MASTERPLAN §5.2).
- A **resumable plan** — the phase breakdown (§13), data model (§3), and risks (§14) can be picked up unchanged once the unlock condition is met.

Explicit do-NOT list while deferred:

- Do **not** create `seller_`, `marketplace_`, `commission_*` tables or migrations.
- Do **not** add `source_kind` / `catalog_listing_id` / `seller_org_id` / `price_lock` fields to the pricing-line Zod schema yet.
- Do **not** scaffold `/commerce`, `/marketplace`, or `/seller/[slug]` routes.
- Do **not** add seller-mode toggles to organizations migrations.
- Do **not** introduce a new top-level nav entry for "Commerce" in `components/shell/nav-rail.tsx` or `components/shell/top-bar.tsx`.
- Do **not** wire the command palette to marketplace actions.

If a Proposal OS task ever appears to *require* one of the above, stop and re-open this document for a deliberate unlock review — do not bypass.

0. North-star principle

One canonical engine. Many tenants. Layered surfaces.

- The proposal/quotation engine (`proposal_pricing` + IR + preset/block registry) stays the single source of commercial truth — for SOL.52, for every vendor, for every storefront.
- Sellers (SOL.52 itself + 3rd-party vendors) are just **organizations with a seller mode enabled**. No parallel tenant model.
- Marketplace, vendor storefronts, commerce workspace are **read/write surfaces** over the same engine. They don't fork pricing.
- Customer-facing experience stays "Shopify-simple" — under-the-hood multi-tenancy is invisible.

This mandate is already locked in MASTERPLAN.md §3 (rule 4), §4, §5 — this plan operationalizes it.

1. Architectural answer to Q7 (the most important question)

Verdict: Layered modular system on a multi-tenant commerce substrate. Not a unified mega-engine. Not three separate silos.

Four discrete layers, each replaceable without breaking the others:

```

flowchart TB
    subgraph L1 [Layer 1 - Canonical Solar Engine]
        SE[solar-engine, tariff-engine, bill-parse]
        PE[proposal_pricing line_items - canonical commercial truth]
        PS[proposal_pricing_snapshots - immutable]
        AE[proposal_approval_events - append only]
    end

    subgraph L2 [Layer 2 - Document Artifacts]
        PROP[Proposal - narrative + savings story]
        QUOT[Quotation - commercial commitment + BOM]
        INV[Invoice - future, post-order]
    end

    subgraph L3 [Layer 3 - Multi-tenant Commerce]
        ORG[organizations - tenant root]
        SELL[seller mode - listings, warehouses, regions]
        CAT[marketplace_listings - SKUs]
        ORD[marketplace_orders + order_items]
        COMM[commission_ledger]
    end

    subgraph L4 [Layer 4 - Experience Surfaces]
        BUILDER[Builder - proposal + quotation]
        HUB[Proposals Hub - pipeline]
        CW[Commerce Workspace - NEW]
        MKT[Marketplace - public discovery]
        STORE[Seller Storefront - public]
    end

    SE --> PE
    PE --> PROP
    PE --> QUOT
    PE -.optional catalog ref. .-> CAT
    ORG --> SELL
    SELL --> CAT
    CAT --> ORD
    ORD --> COMM
    L1 --> BUILDER
    L2 --> HUB
    L3 --> CW
    L3 --> MKT
    L3 --> STORE

```

Why this shape:

- **Unified engine (option A)** would force marketplace into proposal schema → bloated, slow, hard to evolve catalog independently.
- **Three silos (variant of B)** would re-implement pricing logic in catalog/orders → inconsistent totals, snapshot drift, the exact bug pattern we've fought historically.
- **Layered + multi-tenant (the chosen direction)** keeps the engine canonical (no silent forks), lets the commerce layer scale (catalog, inventory, commissions can change independently), and treats artifacts (proposal/quotation/invoice) as **views** over one priced BOM.

Concrete rule that enforces it: every line on every quotation, every order, every commission entry traces back to a `proposal_pricing.line_items` row (or a snapshot of one). No commerce surface inserts pricing directly.

2. Tenant model: one org, many modes

Today an organizations row = an installer company. Tomorrow it can wear up to four **modes** simultaneously, all controlled by additive flags. No parallel tenant types.

Mode	Flag (additive column on organizations)	What it unlocks
Installer (default)	always on	CRM, proposals, projects
Seller / Vendor	marketplace_seller_enabled boolean	listings, warehouses, storefront, commissions
Buyer	marketplace_buyer_enabled boolean (default true)	purchase from marketplace
Platform-owned (SOL.52 itself)	seller_kind enum('partner','platform_owned')	special seller org owned by super admins

SOL.52 as seller (Q1): a real organizations row with `seller_kind='platform_owned'`, owned and operated by super admins. Selling SOL.52-branded panels/inverters is just "this seller is a first-party". No special-casing in the engine.

Customers stay simple: end customers (hotel owners, homeowners) remain rows in `customers/leads` (not orgs). Only businesses that need to *sell* become orgs. This preserves the "low-tech sales person" flow.

3. Data model — additive migrations only

Building strictly on the foundation already shipped in `supabase/migrations/020_organizations_foundation.sql`. Every change is additive (no destructive rewrites, MASTERPLAN §5.2 rule).

3.1 Extend organizations (Phase E1)

```
ALTER TABLE organizations
  ADD COLUMN seller_kind text DEFAULT 'partner' CHECK (seller_kind IN
('partner','platform_owned')),
  ADD COLUMN marketplace_seller_enabled boolean DEFAULT false,
  ADD COLUMN marketplace_buyer_enabled boolean DEFAULT true,
  ADD COLUMN seller_verification_level text DEFAULT 'unverified'
  CHECK (seller_verification_level IN ('unverified','verified','trusted','sol52_verified')),
  ADD COLUMN gst_number text NULL,
  ADD COLUMN legal_name text NULL,
  ADD COLUMN logo_url text NULL,
  ADD COLUMN seller_slug text NULL UNIQUE;
```

3.2 Seller operating tables (Phase E2)

Table	Purpose
seller_warehouses	(id, org_id, name, address, pincode, lat, lng) — per-seller stock locations
seller_delivery_regions	(seller_org_id, state, pincode_prefix, lead_time_days, freight_inr) — where they ship
marketplace_listings	(id, seller_org_id, epc_category, brand, model, wattage, unit_kind, unit_price_inr, warranty_years, datasheet_url, status, moderation_state) — the SKU
seller_inventory	(listing_id, warehouse_id, stock_qty, low_stock_threshold, updated_at) — daily-snapshot stock
marketplace_bundles	(id, owner_org_id, name, listing_ids[], discount_pct, target_system_kw_min, target_system_kw_max) — kits

3.3 Commerce orchestration (Phase E3)

Table	Purpose
marketplace_inquiries	proposal/quote → seller "interested" handshake, lives independently of orders
marketplace_orders	(id, buyer_org_id NULL, buyer_lead_id NULL, status, source_proposal_id, source_quotation_id, total_inr, created_at)
marketplace_order_items	(order_id, listing_id, seller_org_id, warehouse_id, quantity, unit_price_inr_snapshot, commission_inr_snapshot)
commission_rules	(id, seller_org_id NULL, epc_category NULL, percentage, flat_fee_inr, effective_from) — global defaults + per-seller overrides
commission_ledger	append-only entries per marketplace_order_items row, mirrors proposal_approval_events pattern
platform_featured_sellers	super-admin curated feature flags + boost score
platform_recommendation_rules	"if proposal has 5kW residential in Maharashtra, recommend Listing X"

3.4 The single critical extension to the engine

The only change to existing canonical tables is **two nullable columns** on `proposal_pricing.line_items` (already pre-approved in MASTERPLAN §5.2):

```
-- inside the jsonb line_items shape (no SQL migration; schema-level Zod change)
{
  ...existing fields,
  source_kind?: "installer_internal" | "seller_listing",
  catalog_listing_id?: uuid | null,
  seller_org_id?: uuid | null,
  price_lock?: "snapshot" | "live"    // default "snapshot"
}
```

This is the **single seam** that lets the marketplace plug into the canonical engine. Old proposals/quotations keep working unchanged (every field is optional).

4. Pricing ownership — the rule that prevents chaos

Three explicit pricing sources, with **deterministic precedence**:

```
flowchart LR
    L[New line item] --> Q{Source?}
    Q -->|Installer types price| INT[installer_internal - frozen on save]
    Q -->|Picked from marketplace| LIST{price_lock?}
    LIST -->|snapshot - default| SNAP[Copy unit_price_inr at pick time, store catalog_listing_id]
    LIST -->|live| LIVE[Reference listing.unit_price_inr at read time, until quotation status = sent]
    SNAP --> CANON[proposal_pricing.line_items]
    LIVE --> CANON
    INT --> CANON
    CANON --> SNAPSHOT[On status transitions: proposal_pricing_snapshots immutable copy]
```

Rules:

- The moment a quotation is sent, even live lines auto-collapse to snapshot — the customer always sees a stable number.
- Each snapshot remains a row in `proposal_pricing_snapshots` (existing). Audit-grade.
- Approval events log the source change too: "line item X switched from `installer_internal` to `seller_listing`(Waaree 540W)".

This answers **pricing ownership + approval workflow** in one mechanism that already exists.

5. Order routing — keep it boringly simple

```
flowchart LR
    Q[Quotation accepted] --> O[Create marketplace_order]
    O --> SPLIT{Line items}
    SPLIT -->|installer_internal| SKIP[no marketplace order item, installer ships]
    SPLIT -->|seller_listing| ROUTE{Match buyer pincode to seller_delivery_regions}
    ROUTE -->|match found| PICK[Pick nearest seller_warehouse with stock]
    ROUTE -->|no match| FALLBACK[Mark item as needs manual sourcing, notify super admin]
    PICK --> ITEM[marketplace_order_items row with warehouse_id + commission snapshot]
```

No real-time stock juggling for MVP. Daily inventory snapshot + low-stock threshold alerts. Real-time integration is a Phase E6+ concern.

6. Commission system — append-only, never silent

- Default rule: 5% on all marketplace order items (configurable per category by super admin).
- Per-seller overrides allowed (premium sellers may negotiate down).
- On every marketplace_order_items insert: snapshot the commission rule at that moment into the item row (commission_inr_snapshot), and append a commission_ledger entry.
- Ledger never updates — disputes generate new compensating entries (same pattern as proposal_approval_events). This is what auditors and CFOs need.

7. Role permissions — extends existing RBAC

Building on MASTERPLAN.md §3 — three roles already defined: Super Admin / Company Admin / Employee. Add **scoped permissions** within each role; no new role types.

Capability	Super Admin	Company Admin	Employee	Seller Admin*	Customer (read-only via share token)
Approve sellers globally	yes	no	no	no	no
Set global commission defaults	yes	no	no	no	no
Feature a seller	yes	no	no	no	no
Toggle own org's marketplace_seller_enabled	yes	yes	no	n/a	no
Create/edit own listings	yes	yes	per perm	yes	no
Manage own inventory	yes	yes	per perm	yes	no
Create proposal/quotation	yes	yes	yes	yes	no
Convert quote to marketplace order	yes	yes	per perm	yes	no
View own org's commission ledger	yes	yes	no	yes	no
View any org's data	yes	no	no	no	no

*Seller Admin = a sub-permission set on an existing org member with role='company_admin' whose org has seller mode enabled. Not a new top-level role — just a permission expansion.

8. Solar Commerce Workspace (Q8) — the new module

A **new top-level surface** alongside Dashboard, Proposals, Customers, Projects. It's the "trading desk" view of the business.

```

flowchart TB
    CW[Commerce Workspace] --> ROW1
    CW --> ROW2
    CW --> ROW3
    ROW1[Row 1 - Active Deals]
    ROW2[Row 2 - Inbox]
    ROW3[Row 3 - Intelligence]
    ROW1 --> Q1[Live Quotes - sent, awaiting response]
    ROW1 --> P1[Active Proposals - in builder]
    ROW1 --> H1[Hot Leads - high velocity, high health]
    ROW2 --> V1[Vendor Offers - inbound listings matched to my draft quotes]
    ROW2 --> A1[Approval Requests - my approvals + waiting on others]
    ROW2 --> N1[Negotiation Threads - customer counter-offers]
    ROW3 --> I1[Inventory Alerts - my low-stock SKUs if seller]
    ROW3 --> F1[Financing Matches - quotes matched to NBFC offers]
    ROW3 --> AI1[AI Recommendations - next best action per deal]

```

Composition rule: every tile in Commerce Workspace is a **filter over existing data + new commerce tables**. No new "commerce" entities outside what §3 defines. This keeps the surface a *view*, not a silo.

Mobile-first card stack: on phone, three rows become a vertical stack with one card per screen; swipe between sections. Desktop = 3-column grid.

Lives at /commerce (new route). Hub at /proposals stays as-is (deal pipeline). They are siblings, not nested.

9. Seller storefront + marketplace surfaces (Q2, Q3, Q5)

Three public surfaces — all read-only Next.js routes, share-token pattern reused:

Route	Purpose	Note
/marketplace	Public discovery — browse by category/region/brand, featured sellers ribbon	New
/seller/[slug]	Per-seller storefront — brand, listings, ratings, "Get quote" CTA	New, uses <code>organizations.seller_slug</code>
/quote/[token]	Existing pattern from prior plan — quotation public view	Extend to embed branded product cards when <code>source_kind = seller_listing</code>
/proposal/[token]	Existing public proposal view	Unchanged; commerce is <i>additive</i>

Seller-side flow (Q3) reuses existing engines end-to-end:

flowchart LR

```
S[Seller logs in] --> A[Quick Action - Create Quotation]
A --> P{Inventory or external customer?}
P -->|My inventory| B[Builder pre-filtered to my listings]
P -->|External| C[Standard builder + can mix in SOL.52 marketplace items]
B --> D[Branded PDF + web quote + WhatsApp share]
C --> D
D --> E[Customer accepts]
E --> F[marketplace_order auto-created, routed]
F --> G[commission_ledger entry if cross-seller]
```

No new builder. The same /proposal page + the Phase D /quotation builder — with a single new toggle: "Use my inventory only" / "Mix marketplace". Implementation = a WHERE seller_org_id = me filter on the catalog picker.

10. Customer experience — Q5 in one screen

The 30–60 second flow already specified in the prior plan extends with marketplace intelligence — same Cmd-K command palette, same prefill URL pattern:

flowchart LR

```
CMD[Cmd-K - Create proposal for hotel] --> WIZ[Quick Wizard]
WIZ --> O[Org type - Hotel]
WIZ --> K[System size - 50kW]
WIZ --> BR[Preferred brands - Waaree, Adani]
WIZ --> BT[Budget tier - Standard or Premium]
BR --> GEN[Engine generates in less than 30s]
GEN --> P[Proposal - story mode, executive_pitch]
GEN --> Q[Quotation - BOM with branded cards]
GEN --> CMP[Brand comparison view - Waaree vs Adani]
GEN --> WA[WhatsApp deeplink ready]
GEN --> FIN[Financing options - matched NBFCs]
GEN --> UP[Upsell - BESS, AMC, monitoring]
GEN --> MKT[Marketplace CTA - Order through SOL.52]
```

The "intelligence" (financing match, upsell, brand comparison) is a **service layer** call (lib/ai-quick-suggest.ts from prior plan, extended) — not new entities. Output renders into existing proposal blocks via the block registry — no new render pipeline.

11. Simplicity manifesto (Q9) — non-negotiable rules

These become enforced UX laws in the design system:

1. **No screen with more than 5 required fields.** Always defaults from org type + region + brand prefs.

2. **WhatsApp is the primary delivery channel** for residential & small commercial. Email is secondary. PDF is on-demand.
3. **One word labels.** "Items" not "BOM". "Total" not "Net Cost". "Send" not "Dispatch". Hindi parity for every label via useLanguage.
4. **AI fills the unknown.** Every form field has an "I don't know" / "Suggest" affordance backed by ai-quick-suggest.
5. **Mobile is the design floor, not the ceiling.** Every new surface designs phone first, scales up.
6. **No nested modals.** Drawers and full-screen sheets only on mobile.
7. **Quick Actions reach everywhere.** Cmd-K from any screen creates anything.

These rules go into `lib/design-system.ts` as comments + a checklist in `docs/SIMPLICITY_LAWS.md` (new doc).

12. What evolves vs preserves (Q10)

Preserve (do not touch)

- solar-engine, tariff-engine, bill-parse — calculation truth
- proposal_pricing line shape (only additive optional fields)
- proposal_pricing_snapshots, proposal_approval_events — audit pattern
- WEB_RENDERER_REGISTRY, lib/proposal-block-registry.ts, lib/proposal-preset-engine.ts — block IDs are stable
- Public share token model (/proposal/[token])
- Residential vs commercial renderer split — existing WebRenderer + CommercialProposalView
- E1 design tokens + E2 OsShell + E3 Proposals Hub — just shipped, don't refactor

Evolve additively

- organizations — add seller mode flags (§3.1)
- proposal_pricing.line_items Zod schema — add optional source_kind, catalog_listing_id, seller_org_id, price_lock
- Builder catalog picker (existing) — gains a "Marketplace" tab next to "My catalog"
- Command palette (components/shell/command-palette.tsx) — gains marketplace search + seller search
- OsShell breadcrumbs — gain "Seller > Listings" trail when in seller mode

Become modular

- Commerce surfaces — new `app/(main)/commerce/` and `app/marketplace/` route groups, isolated from `proposal/*`
- Seller settings — new `app/(main)/settings/seller/*`

Become AI-assisted

- Brand recommendation per region/org type
- Quotation drafting from voice/text prompt
- Upsell card on every accepted quote
- Inventory low-stock natural-language alerts ("Waaree 540W will run out in 6 days at current pace")

Become marketplace-ready

- Every `epc-component-catalog` category already has `marketplaceReady` flag — listings table reads it as a constraint
- Every `proposal_pricing.line_items` row already supports `catalog_category` — bridge column to `marketplace_listings.epc_category`

Newly workspace-driven

- "Active workspace" pill (E2) extends to active **commerce context** — current quote, current order, current listing draft
- Approval feed + commission feed appear inside Commerce Workspace, not buried in settings

13. Phase E roadmap (sequenced, additive, shippable)

Phase D (Instant Proposal + Quotation) is the prerequisite — already planned, partly underway. Phase E begins after Phase D ships D0–D3.

Phase	Scope	Ship-gate
E1 — Seller foundation	Org table flag migration + Seller settings page + super admin seller-approval queue	One installer org can flip seller mode on
E2 — Listings + storefront	<code>marketplace_listings</code> + <code>seller_warehouses</code> + <code>seller_delivery_regions</code> + <code>/seller/[slug]</code> public page + listing CRUD UI	A seller can publish a listing visible at <code>/seller/their-slug</code>
E3 — Catalog plug-in to engine	Add <code>source_kind</code> / <code>catalog_listing_id</code> to pricing-line Zod + Builder catalog picker "Marketplace" tab + snapshot logic	An installer's quotation can pull a marketplace listing as a line item with proper price snapshot
E4 — Public marketplace + recommendations	<code>/marketplace</code> discovery + <code>platform_featured_sellers</code> + <code>platform_recommendation_rules</code> + featured ribbon on builder	A buyer can browse, compare, and add to quote
E5 — Orders + commissions	<code>marketplace_orders</code> + <code>marketplace_order_items</code> + <code>commission_ledger</code> + <code>accept→order</code> flow + super admin commission console	Accepted quote auto-creates routed order with commission entry
E6 — Commerce Workspace	<code>/commerce</code> route + tiles (live quotes, vendor offers, inventory alerts, financing matches, AI recs)	Mobile + desktop dashboards live
E7 — Inventory automation + verification badges	Inventory snapshot jobs + low-stock alerts + SOL52 Verified Vendor badge (<code>seller_verification_level</code>) on storefront + builder	Sellers get inventory pressure visibility; buyers see trust signals

Each phase is independently shippable. No phase blocks a customer-visible feature for more than 2 weeks of work.

14. Risks & explicit non-goals

Risks watched

- Pricing drift between snapshot and live listings → mitigated by auto-collapse to snapshot on sent (\$4)
- Commission disputes → mitigated by append-only ledger (\$6)
- Seller spam → mitigated by super admin approval gate + verification levels (\$7, E1)
- "ERP creep" — too many fields, too many screens → mitigated by simplicity manifesto (\$11) enforced in code review

Non-goals (deferred)

- Payment processing / settlement — Phase F+
 - Multi-currency — Phase F+
 - Real-time stock sync with vendor ERPs — Phase F+
 - Logistics integration (Delhivery/Shadowfax APIs) — Phase F+
 - Buyer-side org accounts — Phase E8+ (customers stay as leads/customers for now)
-

15. Open questions to confirm before E1 begins

These will be raised when we move from plan to execution. No need to answer now:

1. Should SOL52 platform-owned seller org be seeded as a real migration row, or created via super-admin UI in E1?
2. Default commission percentage at launch — 3%? 5%? Per-category split?
3. Verification badge tiers — keep 4 (unverified → sol52_verified) or simplify to 3?
4. /marketplace discovery — search-first (IndiaMART style) or curated-first (Shopify-collections style) at launch?
5. Cross-org employee invitations — out of scope for E1, or do we need read-only "customer" portal accounts before E5?